ZOTGPT

G: Gerbil
P: Portable
T: Transportation

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ENGR 7B: Introduction to Engineering

University of California, Irvine

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Table of Contents (Required)

Executive Summary

When we first started the project we wanted to set some goals so that we could be on the same page at the start. Our goal was to communicate better, split the tasks evenly, and improve our design and coding skills. We held our meetings at the science library on Thursdays from 2 pm to 3 pm. On our first meeting, we decided our roles and tasks, and that we would want a lightweight rover that is one layer and is fast. To tackle the lightweight part we decided to get the ¼ birch plywood as it was the best mix of stability and lightness. For the rover to be fast we decided to get the 34:1 motor with the big wheel as it'll allow for twice the distance to be covered per rotation. For the initial IR sensor placement, we thought it would be good to put one in the back and two in the front, however after discussing with our TA, we had to put the IR sensors in a row close to each other. To make wiring less complicated we put the IR sensors in the middle of the chassis in line with the motors, which would become a problem later on. Since the IR sensors were in line with the motors there was not enough time for the IR sensors to process and send a signal to the motors, which caused a lot of stuttering and it wasn't following the line correctly. To overcome this we drilled holes in the front and put the IR sensors there. The CAD team would then get to work designing the chassis going on the side of caution when choosing the dimensions to allow for all the electronics to fit comfortably on the chassis. This would turn out to not be an issue and the chassis could have been reduced in size as there was some excess. During the fabrication process, the chassis was laser cut and the wiring and coding began to be done. The size and holes in our chassis allowed for roots of freedom when placing electronics, with little to no issues coming along. The most important part of wiring was making sure that the buck converter and micro servo were properly adjusted so as to not draw excess power to the micro servo. While coding our team started with a simple pseudo-code outline of what needed to be accomplished by the rover. After many syntax and logic errors, we were able to get our rover to line track, and separately we adjusted our rover's pixy cam to properly detect the color of the can. While putting both codes together we ran into issues of the rover prematurely detecting the can but this was later fixed. In the end our rover was able to complete the course and finish in the top 18 of groups.

Problem Definition

Introduction

The objective of this course was to design and create an autonomous rover. This rover had to be able to navigate through a course using line tracking and then afterward pick up a colored object using a claw we designed as quickly as it could. To fulfill this objective, our team optimized our rover for speed by designing the rover around its weight and motor/wheel options. Coming into this quarter, our team realized that our main obstacle was going to be the electrical and coding part of our rover as they were the items that were going to be built on in this quarter. Our goal was to brainstorm and finalize our design and CAD for our rover as quickly as possible which would allow us more time to tweak and improve our code.

Technical Review / Background

Rovers have had an interesting history, progressing from their first space research missions to becoming essential parts of contemporary technology. Since NASA's groundbreaking planetary exploration missions, which were led by the Mars rovers Spirit and Opportunity, rovers have been essential in expanding our knowledge of far-off places. These autonomous vehicles have evolved beyond space exploration throughout time, finding use in a variety of industries including research, logistics, and defense. Robotics and artificial intelligence developments in the modern era have elevated rover technology to unprecedented levels. Sophisticated autonomous navigation systems, resilient sensor arrays for environment awareness, improved communication capabilities, and even food delivery robots that can be seen on our very own UCI's campus are examples of current advances.

During our research and brainstorming, we decided on a few major design details. Those included a flat, one-layer chassis, two 80mm wheels, and one caster wheel, and putting the PixyCam on top. These design choices resemble many modern rovers, especially those of the rovers that NASA put on Mars. We believed the wide and one-layer chassis could help provide a balance between stability and lightness for speed. Additionally, the PixyCam that we raised with a custom 3D-printed part works similarly to the ones on the Mars rovers as it allows it to see more with its higher vantage point. This could work to our advantage when our rover was tasked to grab a colored can.

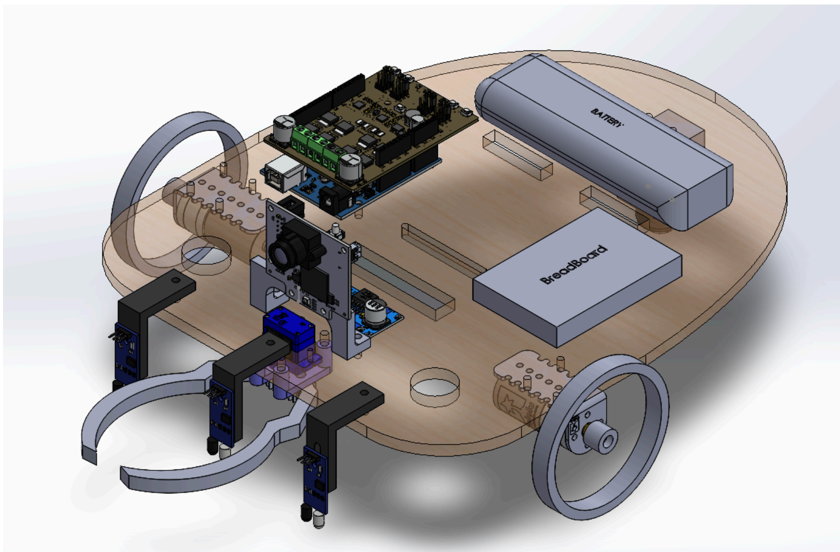
Design Requirements

1. Must be smaller than 12x16
2. Materials cost less than $300
3. Must be safe (no sharp objects protruding out)
4. Well insulated wiring
5. Must be able to grab a can
6. Must be Autonomous
7. Adequate weight to prevent stalling
8. Battery must be easily accessible

Design Description

Summary of Design

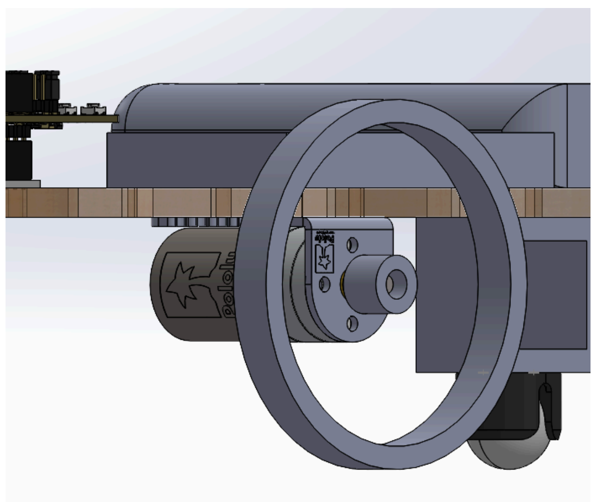
In our rover, we decided on a beetle-like design as we saw that it would be beneficial to base our rover on something from the real world. The IR sensors are placed directly in the front so that they detect the line before activating the motors. We placed the motors more in the front because we wanted the rover to turn with the wheels instead of the caster ball. The battery is placed in the back for better weight distribution and is fastened by two velcro straps wrapped around the rectangle cutouts.. Everything is mounted on M3 screws of various lengths. In addition, our rover would be done in one layer.



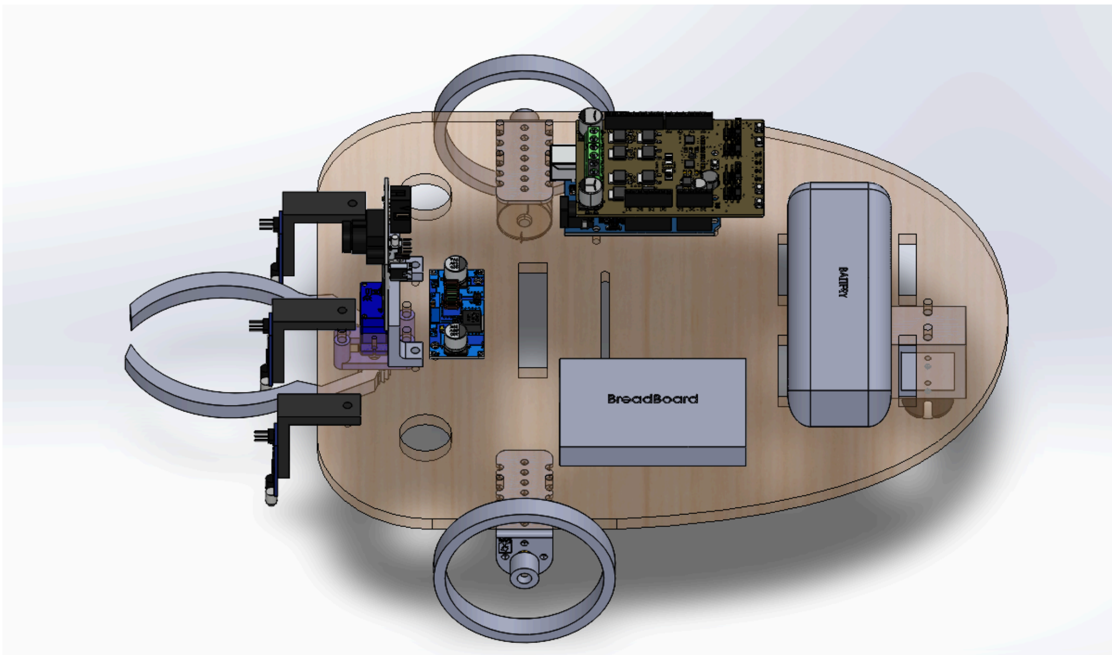
Our total weight for our rover is 1,068.9 grams and the total cost is $263.41 with a length of 15.25in by 11.27in.

Design Details

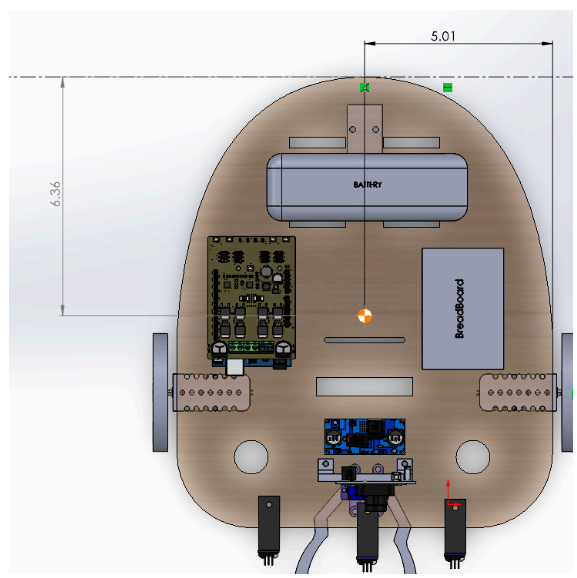
Our rover consists of a chassis made of birch plywood that is lightweight and sturdy to support our electronics and hardware. The motors for this rover are the 34:1 motors with the largest wheel offered to us so that we get a fast-moving rover. This is because with a fast motor and a bigger wheel, it is able to move more distance because of the circumference of the wheel.



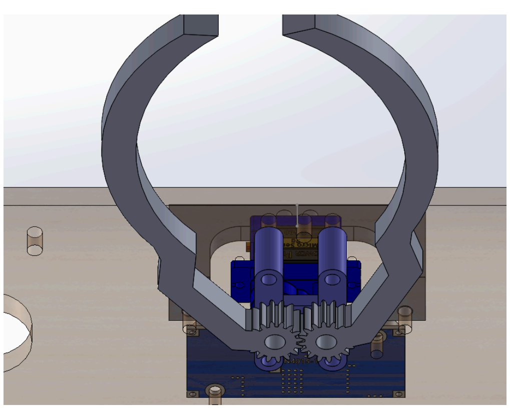
Since our chassis is 12in by 10in, which is larger than most rovers, we had to push the claw a little bit back into the chassis to be within the 16in by 12in constraint dimension of the project.



The IR sensors are placed within the front and are mounted by L-brackets that hang off the sides so that they are secure and get as close to the ground as possible. The motors are placed under the chassis so that there would be more space on top for the electronics and wiring to keep things organized. The claw and IR sensors might seem to interfere with each other but the code allows it so that would never happen.

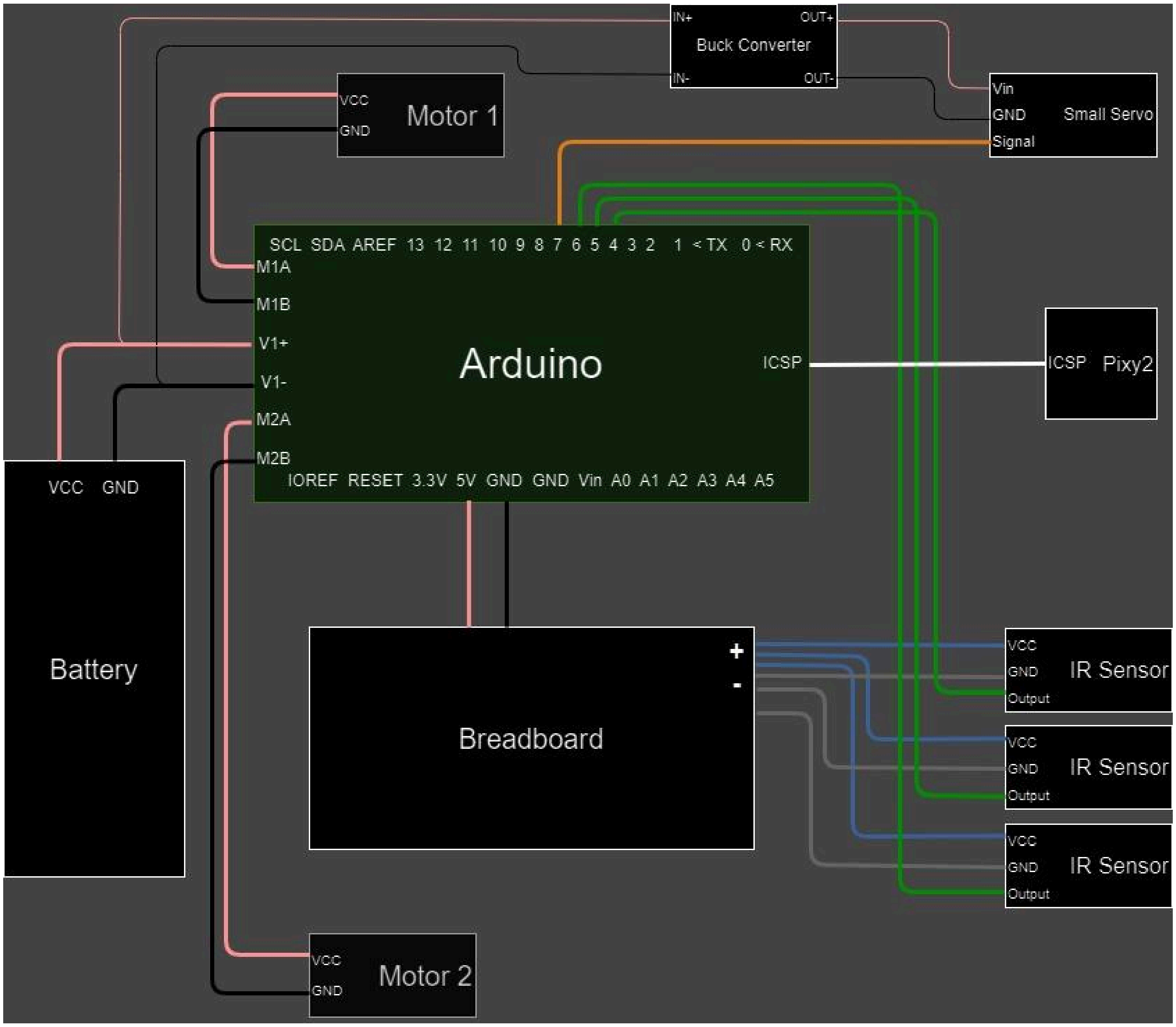


The center of mass of our entire rover is 5.01in from the widest part of the width and 6.36in from the back of the rover.



The claw of our rover is fixed to a 3d part that holds both the claws together. We chose this design of the claw because it is simple and easy to 3D print. The claw didn't have to be complicated for it to fit the criteria of grabbing a can.

Wiring Diagram

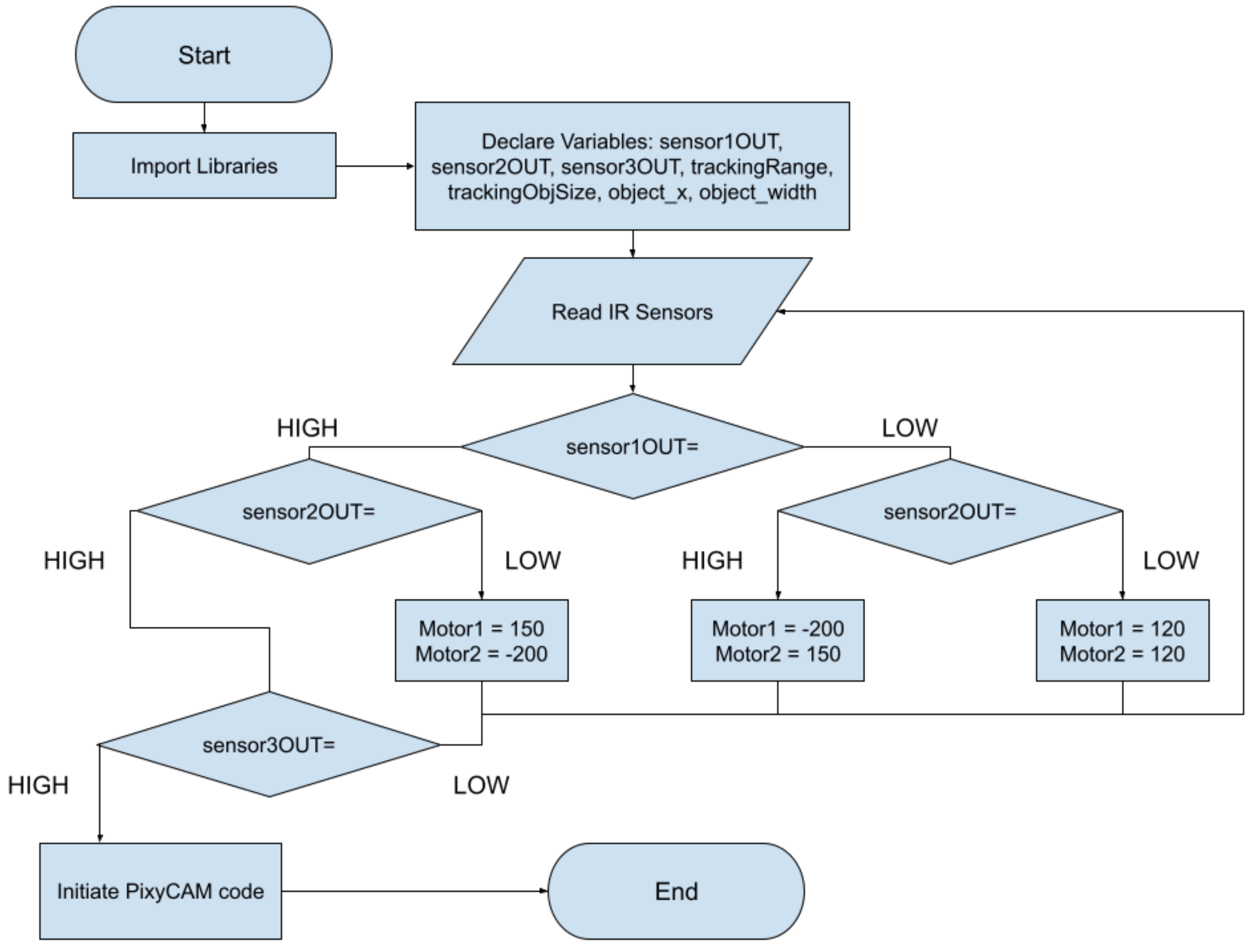


Algorithm Design

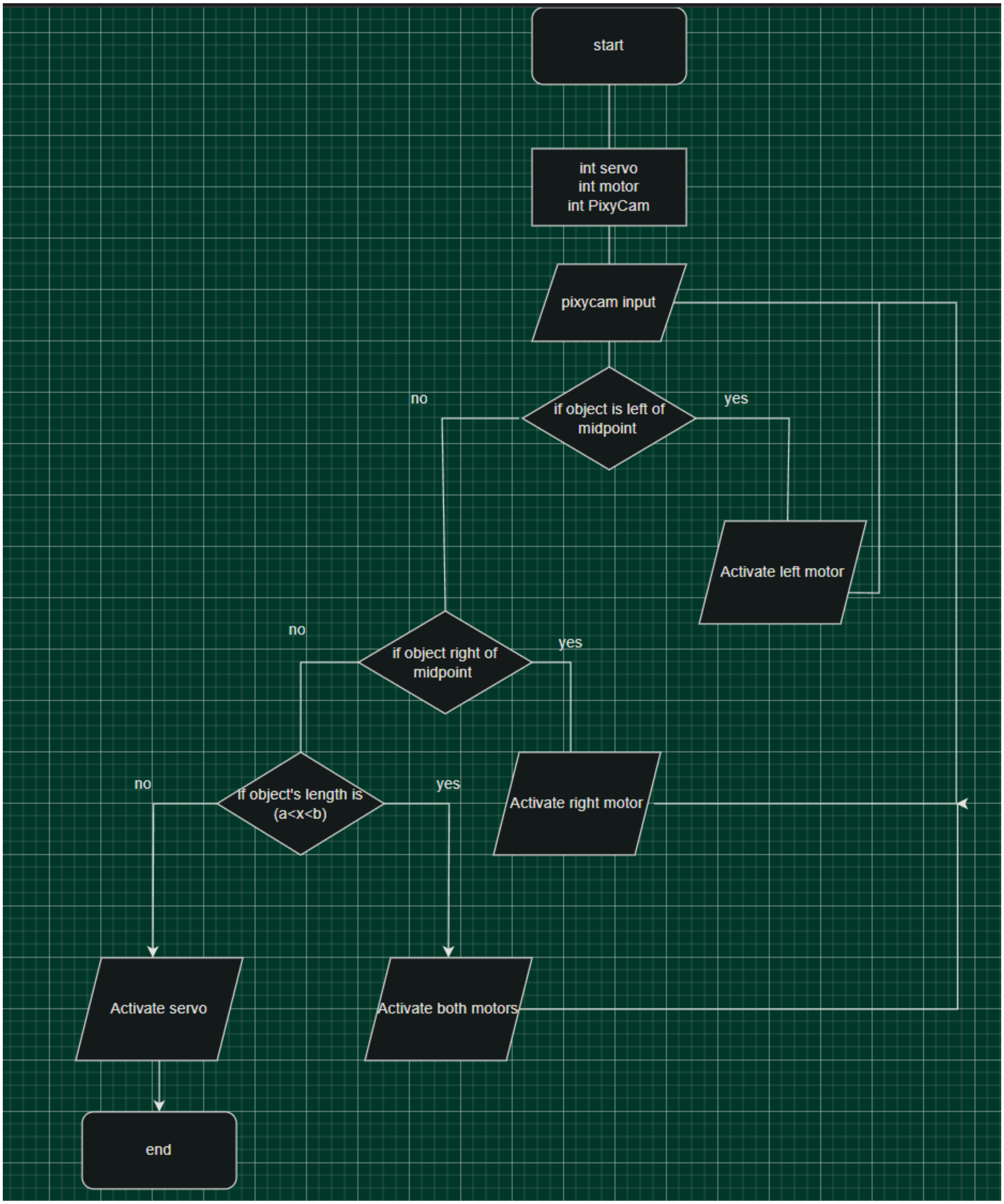
The code consists of two main sections, the line tracking and PixyCAM algorithms, in the form of two user defined functions. The line tracking algorithm relies on three IR Sensors. Two IR sensors are used to self-correct its direction based on when the sensors detect the line. There are three main cases: When detecting the line on the left sensor, it will turn right, and vice versa; when detecting nothing on the left or right sensors, it will go straight; and when all three sensors detect the line, it will switch over to the PixyCAM code. The PixyCAM algorithm tracks the position of an object and turns the rover until it is completely centered in the camera's field of view. Then, the rover moves forward until the camera sees the object as a certain threshold size and closes the claw.

Pseudocode:

Line Following Flowchart:



PixyCAM Flowchart:

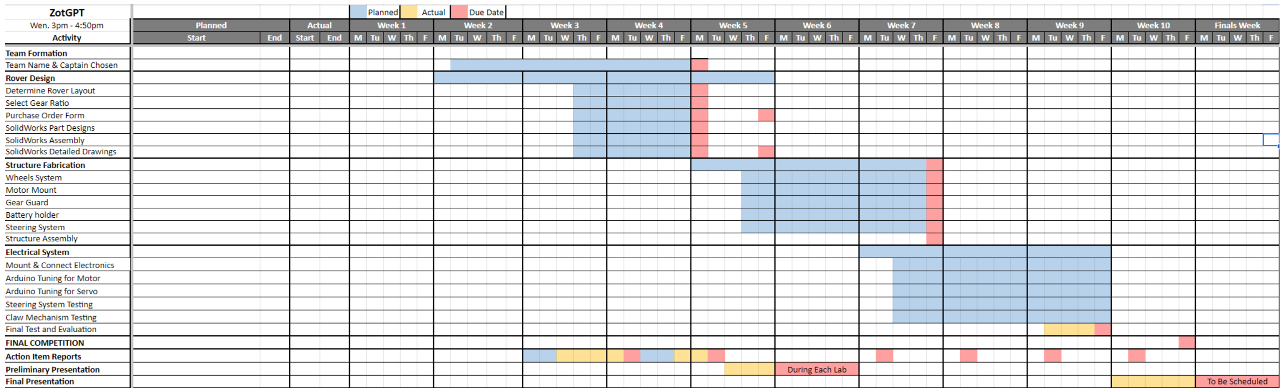


Action Item Report

Task Assignment

At the beginning of the class, we first took the time to see who was going to take on which roles. FIrstly, Bryant was elected as team captain. With this role, he would keep the team on track to completing our tasks in a timely manner. Under the CAD team, we had Bryant, Ben, and Maddox. The CAD team would take the ideas the team had at the beginning of the project and create models of them on Solidworks for testing and construction in the future. Next, Bryant and Ben were once again on the fabrication team where they would take the models from Solidworks and use them to create the physical components of the rover. Once everything was made, such as the chassis and the mounts for the various sensors, they took the individual parts of the rover and put them together. The next step was to put together the electronics for the rover, which Cole, Maddox, and Rafael were responsible for. In order for the rover to successfully detect the lines, drive, and then finally grab the object at the end of the course, a variety of electronics were needed. The electronics team made sure to mount the necessary devices to the chassis of the rover, then made all the connections needed to ensure every component had power and drove as expected. Some of these components included the Arduino, motor shield, breadboard, motors, and wires to connect all these pieces. Finally was the coding team, which Cole and Rafael were a part of. The Arduino required coding to let the rover know where to go. Part of the coding involved detecting where the black line it needed to follow was, where to turn, and where to drive to pick up the object at the end of the course. Cole and Rafael were involved in the coding team to accomplish this goal. It required lots of trial and error to get the code to do its task, but it finally was finished and accomplished the goal it needed. Overall, the roles were evenly distributed amongst all the team members, and as the quarter progressed, everyone became accustomed to their tasks and the team ran fluidly.

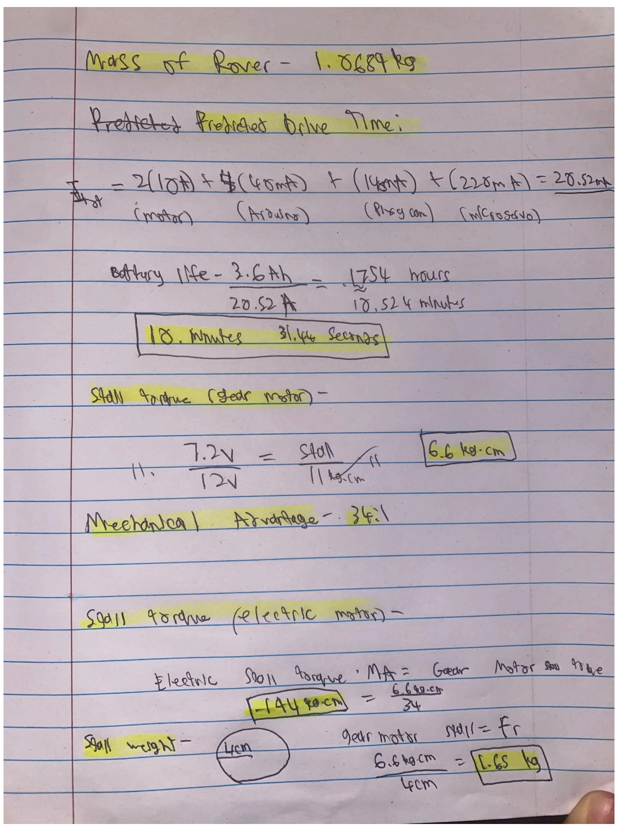
Gantt Chart

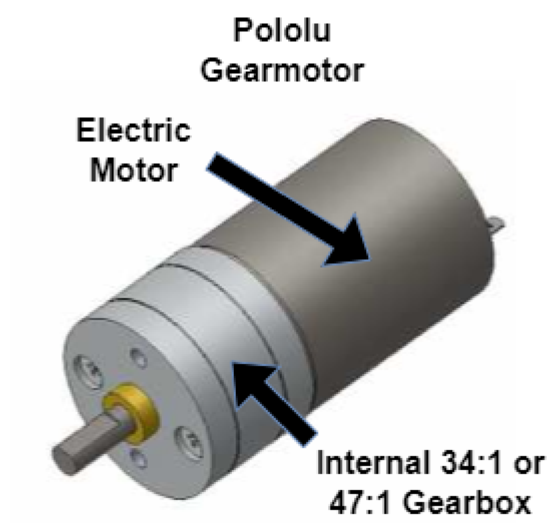


Evaluation

Calculations

The rover's total weight of 1,068.9 grams is less than the stall weight of 1,650 grams which allowed our group to opt for a 34:1 gearbox motor to prioritize speed rather than torque but still giving a mechanical advantage of 34.





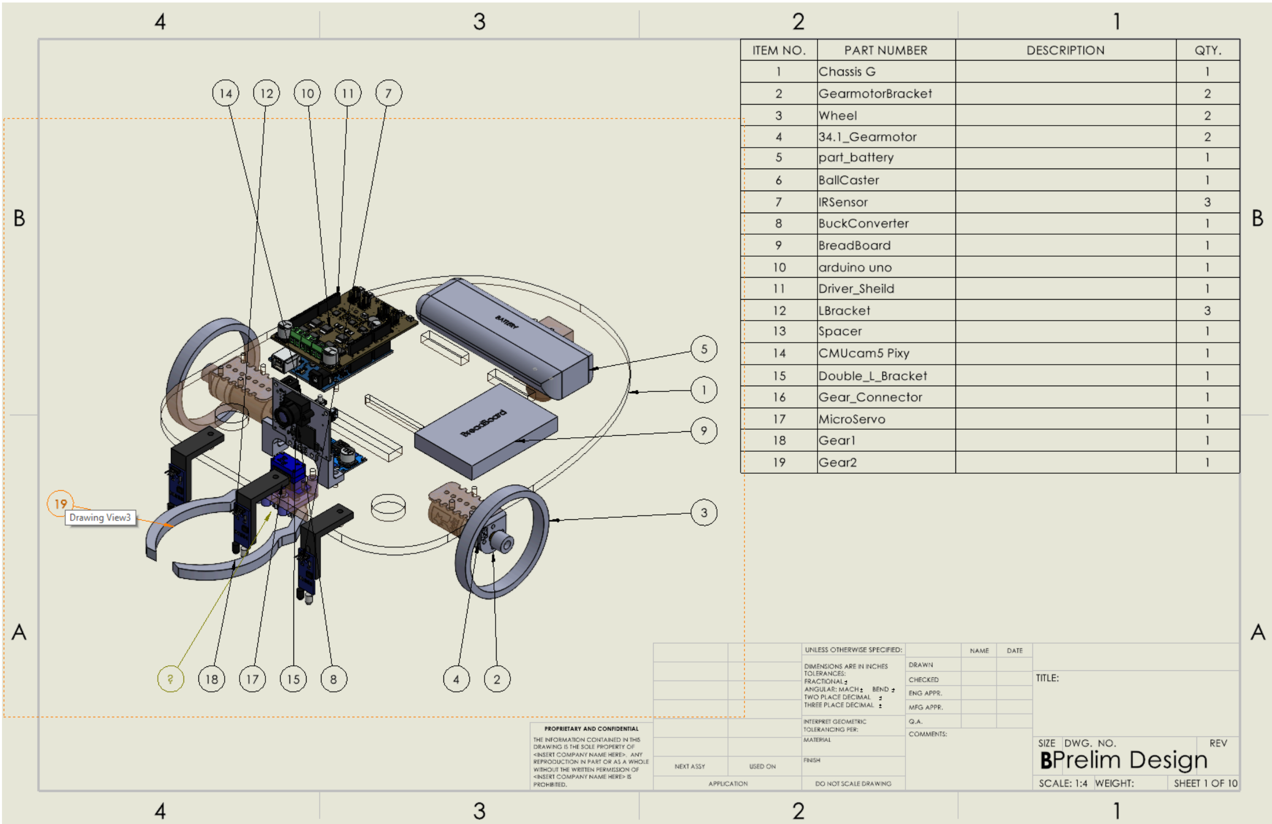
Test Plan

To test our rover we first tested the code for the motors to see if they worked properly, then we tested how the IR sensors reacted with the motors. When the IR sensor test was successful we moved on to test the line tracking and observe if there were any ways we could improve. After the line tracking was successful we tested how the motors would react to the pixycam and then how the servo would react to the pixycam.

Results & Discussion

When we tested our rover we had a lot of trouble at the start of trying to figure out how the motors work and how the IR sensors work. At first, there were some manufacturing issues that were difficult to fix due to the difficulty of putting on the wheel. Because of this it slowed our project down and the wheel was damaged in the process. However, after testing in code we figured it would be a hardware problem with how the IR sensors were placed. So, we placed the IR sensors more in the front and closer together, which helped the code read the line. However, after that problem was fixed we found out from testing on the final track that our rover was too wide and it would get stuck at the edge of the platform. For this problem, there weren’t any solutions due to the time constraints so we had to rely on luck so that the rover would not get stuck on it. When testing the Pixycam it was mostly just a problem in the color of the can, since the light inside the lab was different from the light where the final track was located. To solve this problem, we set the signature to the can with the light at the final track instead of the one inside. If we were to do this project again we would have a smaller chassis and put the IR sensors in front of the motors/wheels and close to the ground.

Appendix A: SOLIDWORKS Drawings (Includes BOM)



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Chassis G		1
2	GearmotorBracket		2
3	Wheel		2
4	34.1_Gearmotor		2
5	part_battery		1
6	BallCaster		1
7	IRSensor		3
8	BuckConverter		1
9	BreadBoard		1
10	arduino uno		1
11	Driver_Sheild		1
12	LBracket		3
13	Spacer		1
14	CMUcam5 Pixy		1
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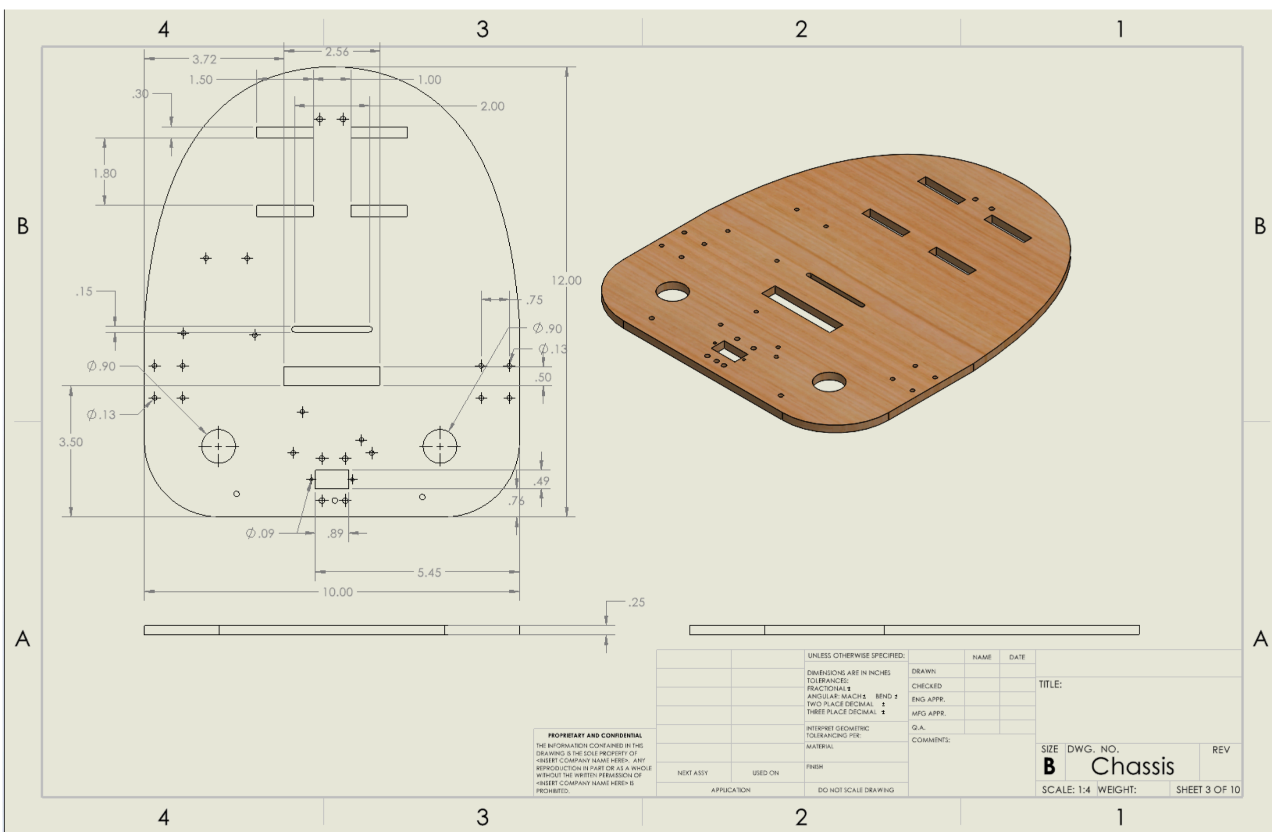
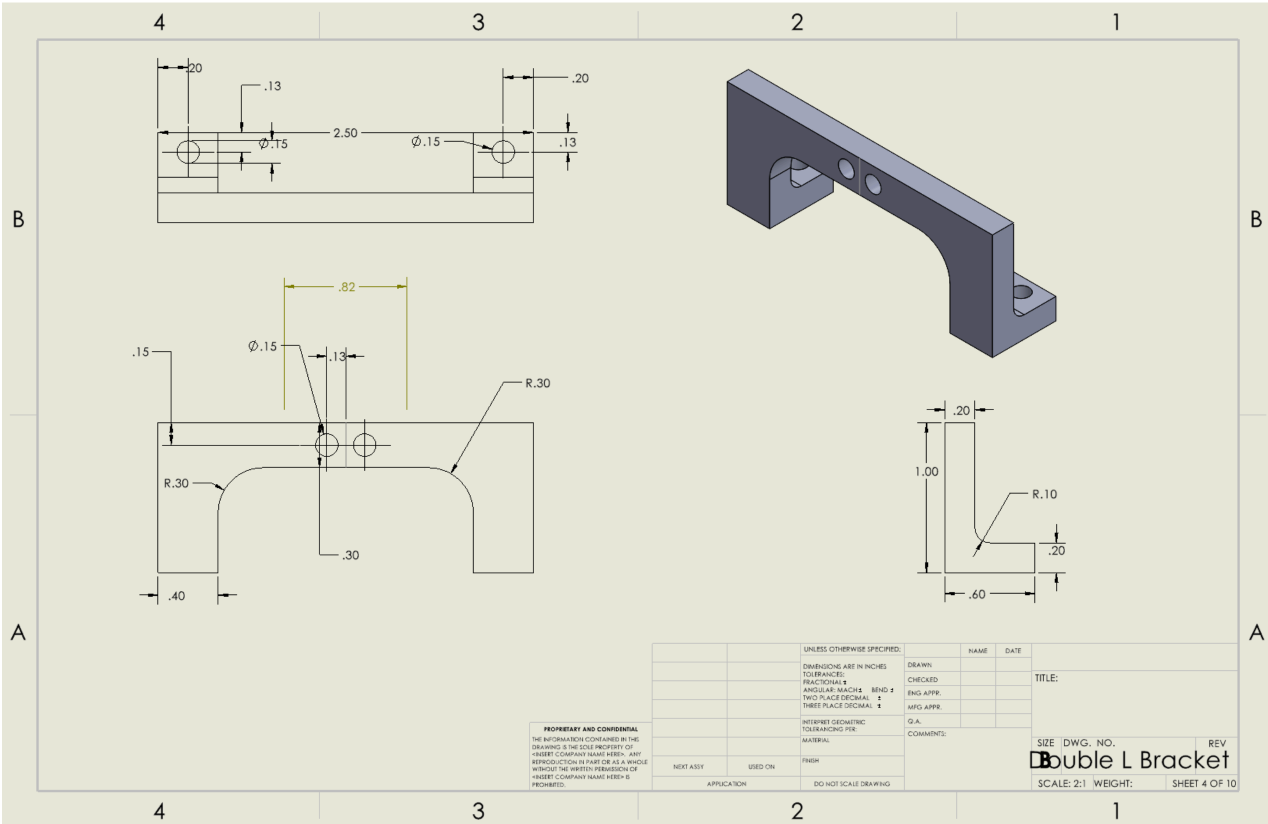
16	Gear_Connector		1
17	MicroServo		1
18	Gear1		1
19	Gear2		1

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Ø.15
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.13
R.30
R.30
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.60
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NAME
DATE
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COMMENTS:
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Double L Bracket
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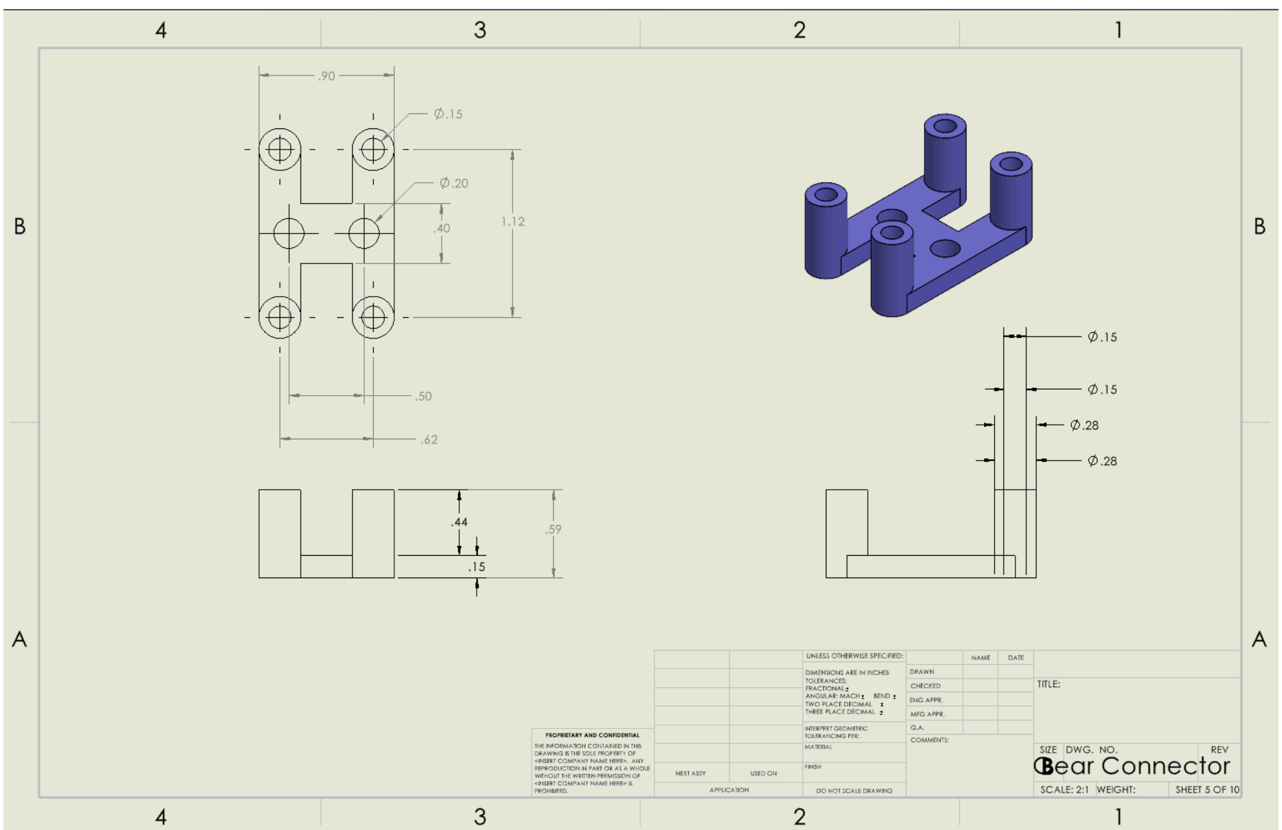
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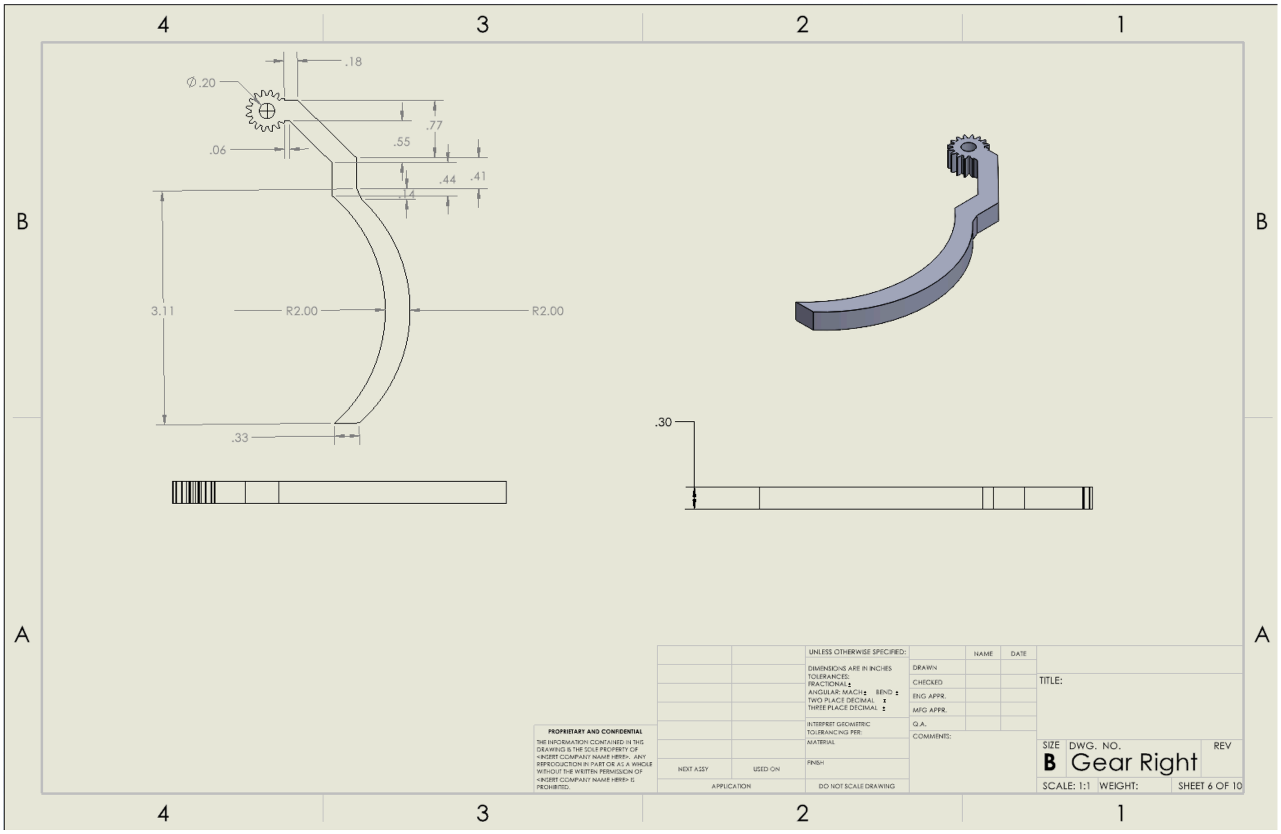

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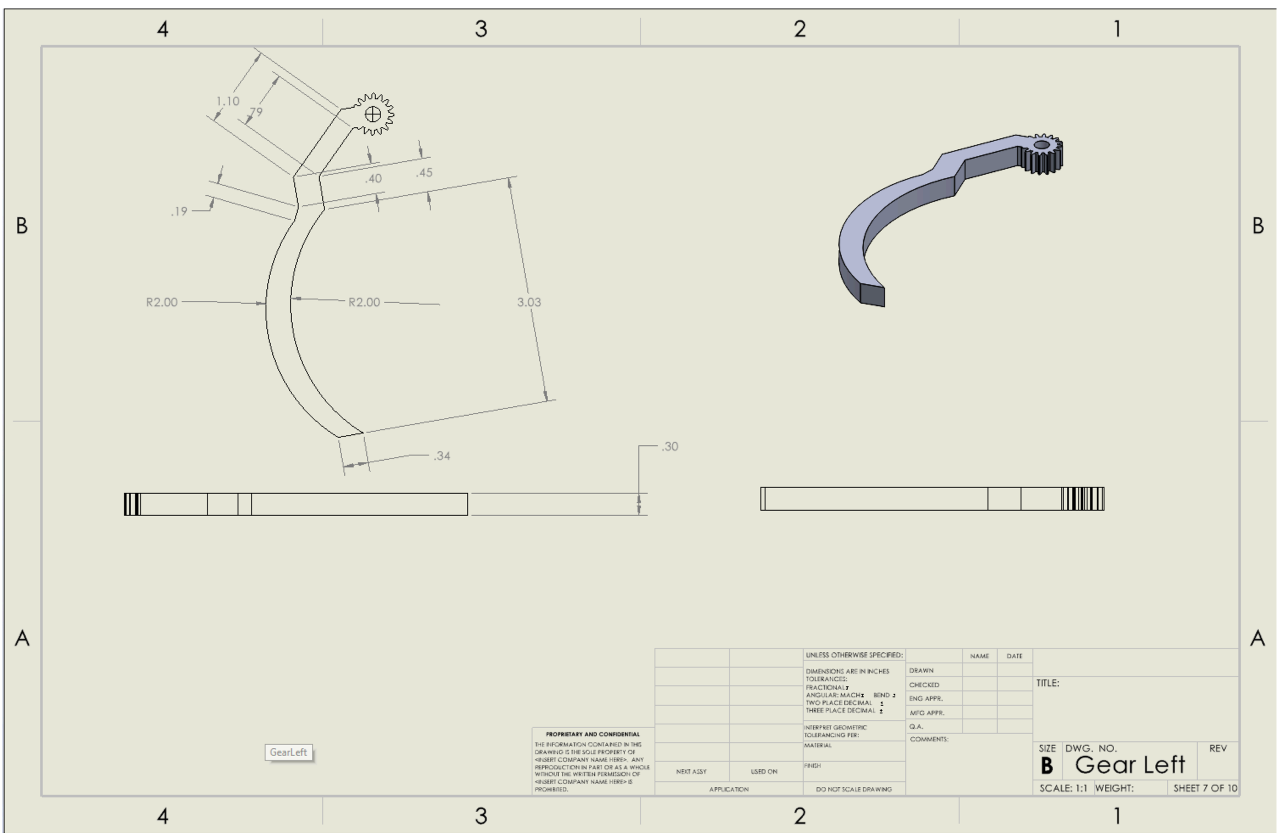

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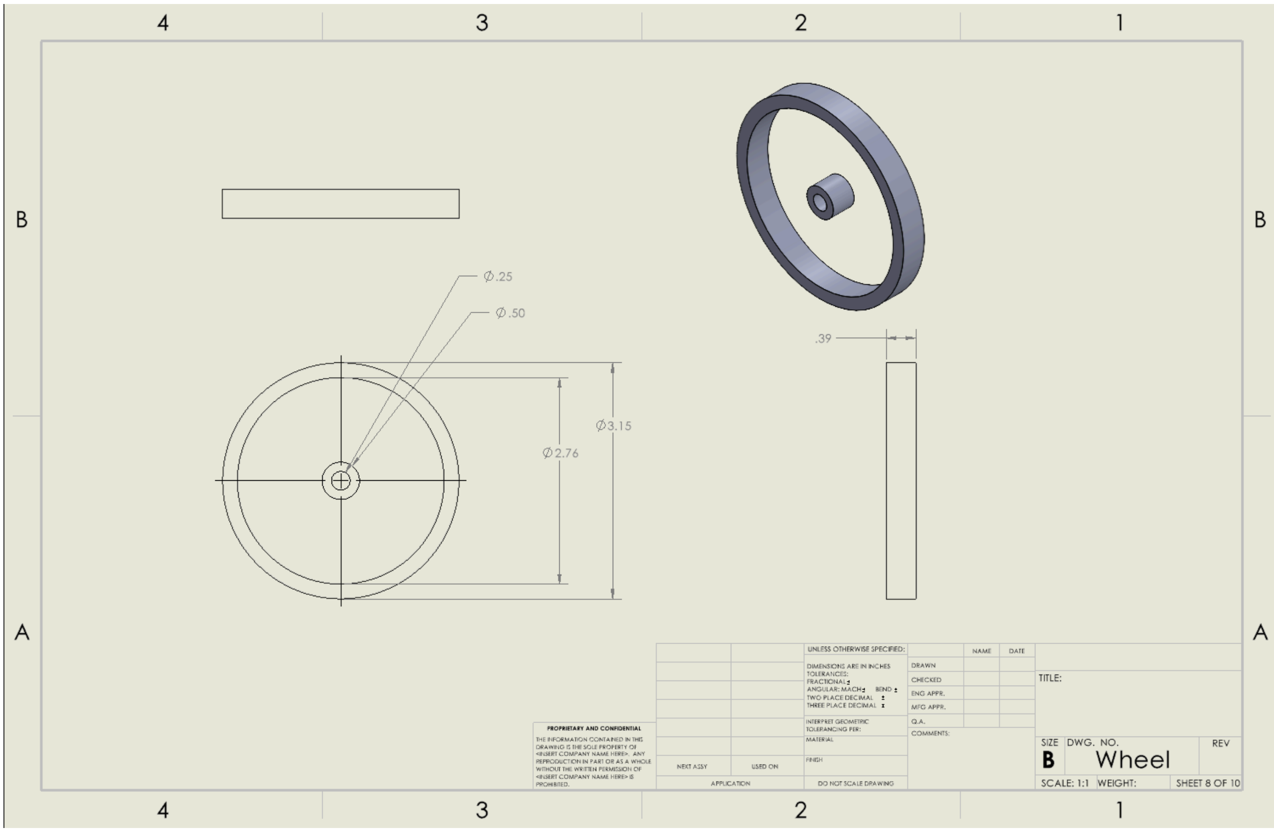
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ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
FINISH
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DATE
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MFG APPR.
Q.A.
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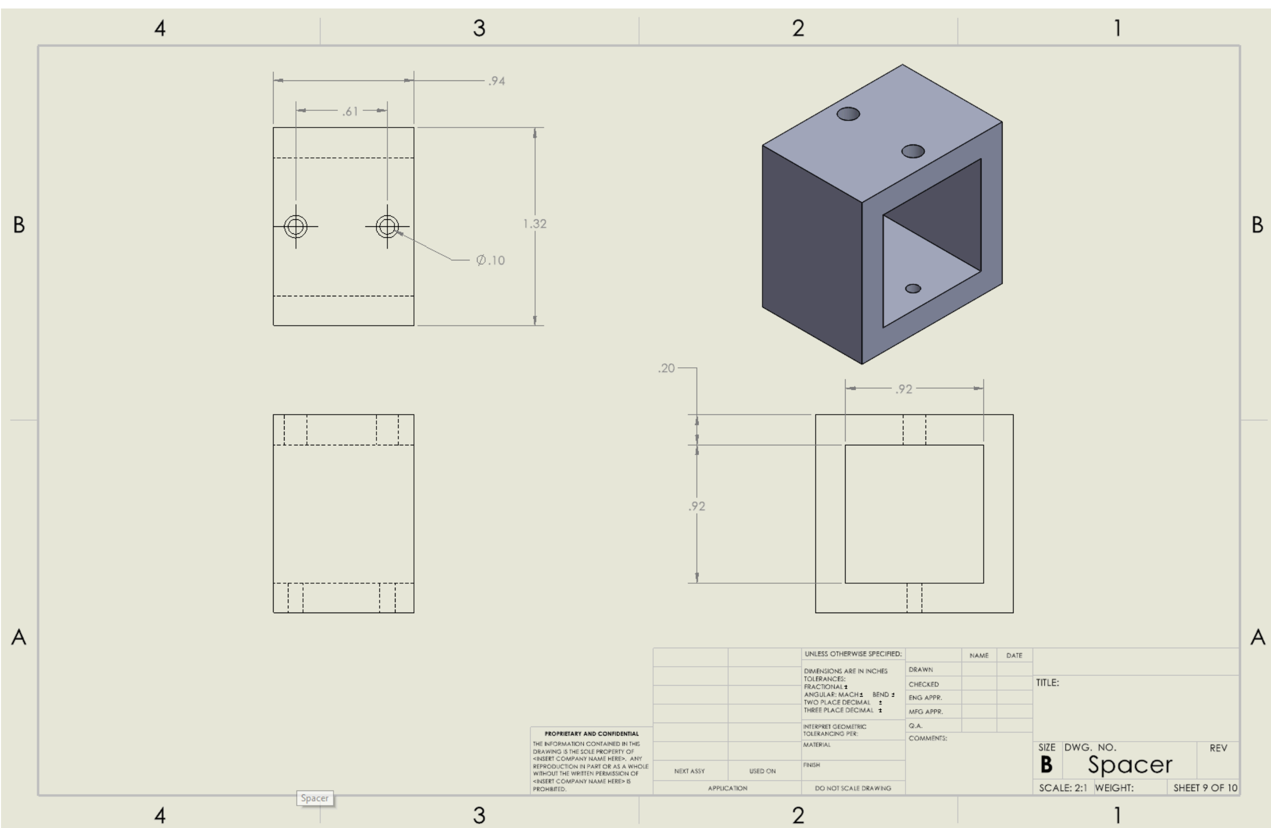
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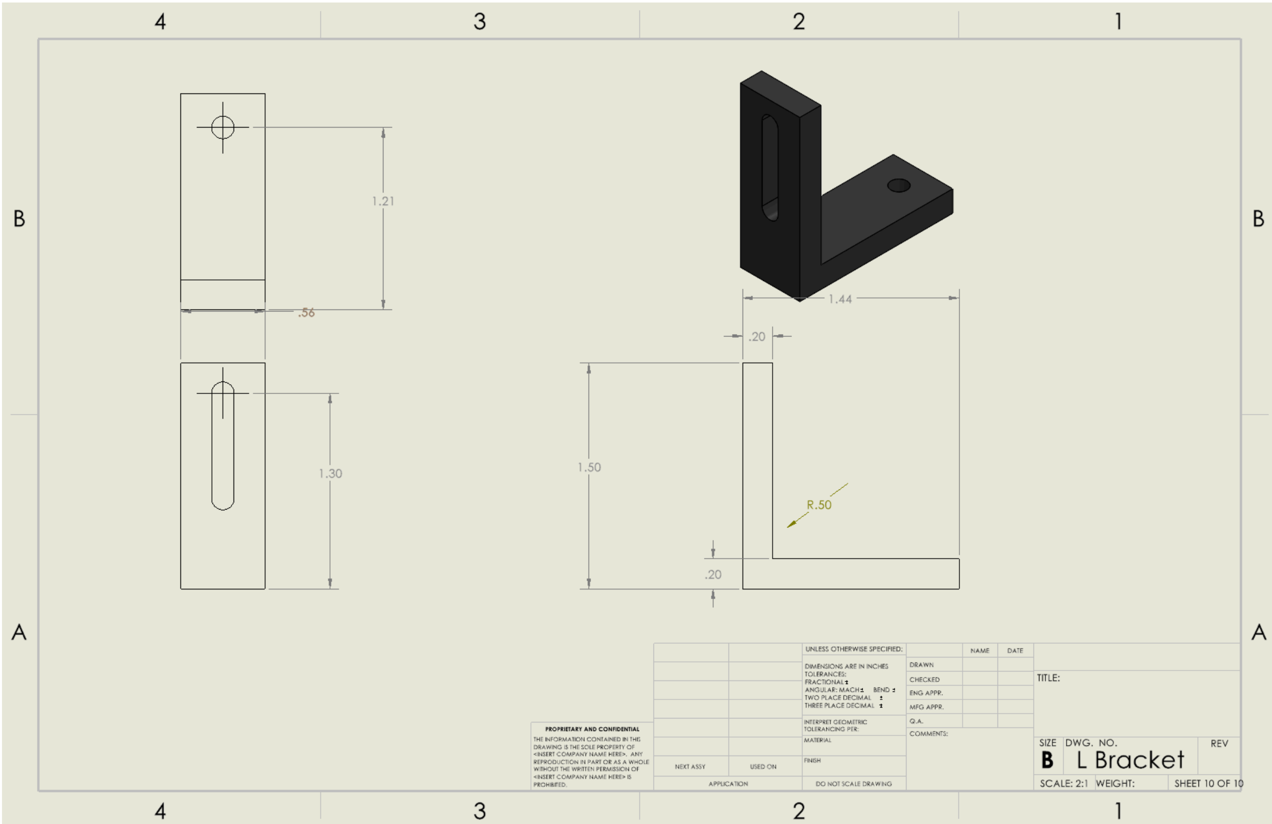
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ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
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MATERIAL
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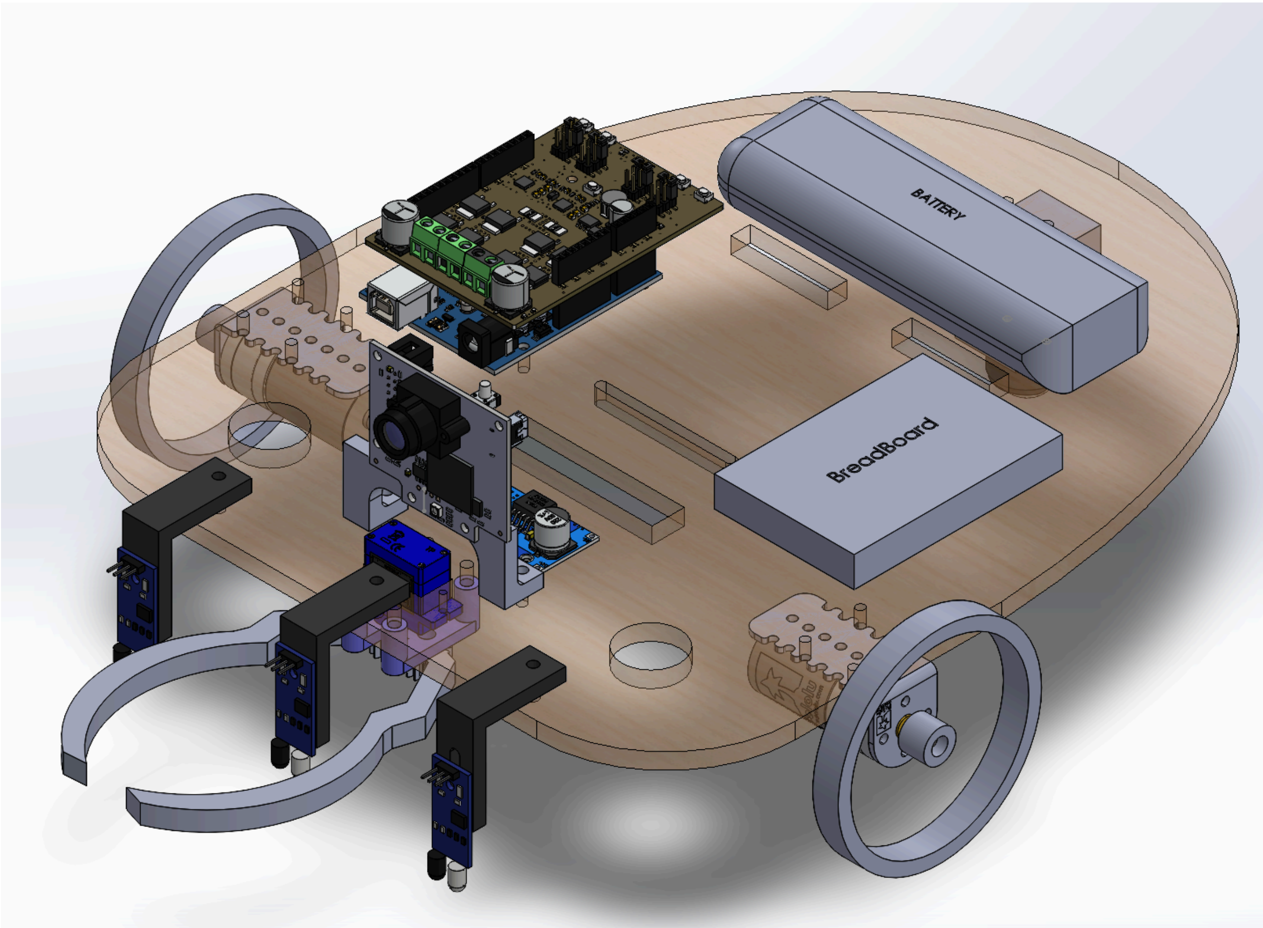
BATTERY
BreadBoard

Appendix B: Bill of Materials

Quantity*	# of Units in package*	Company*	Item Description*	Catalog #*	Price*	Estimated Extended Price*
1	1		Ardunio Uno	1	$27.60	$27.60
1	1		Motor Driver Shield	1	$22.55	$22.55
1	4	D-Planet	Buck Converter	1	$14.99	$3.75
1	6	DEYUE	DEYUE breadboard Set Prototype Board - 6 PCS 40	1	$8.99	$1.50
1	3	Tamiya	Battery Connectors	1	$8.69	$2.90
1	10	SIXQJZML	Power switch	1	$7.99	$0.80
1	2	Zeee	Battery	1	$34.99	$17.50
1	1	Pixy	PixyCam2	1	$69.96	$69.96
1	20	Frienda	IR sensor	3	$10.99	$0.55
1	12	SG90	Microservo	1	$20.99	$1.75
1	1	TYUMEN	Wire	1	$16.99	$16.99
1	1	Pololu	34:1 Gearmotor	1	$29.95	$29.95
1	1	Pololu	80mm x 10mm Wheels	2	$9.95	$9.95
1	1	Pololu	3/4" Caster wheel	1	$3.95	$3.95
1	1		3D printing		$32.00	$32.00
1	1		1/4" birch baltic blywood 16"x10"	1	$2.78	$2.78
					Subtotal	$244.47
				*Tax rate:	7.75%	$18.95
					TOTAL ORDER PRICE	$263.41

Appendix C: Arduino Code

Include full code from Arduino, use comments within code for added clarity.

Appendix D: References

Works Cited:

"Every Mission to Mars Ever." *The Planetary Society*, www.planetary.org/space-missions/every-mars-mission.

Staff, The Robot Report, et al. "The Robot Report." *The Robot Report*, 18 Mar. 2024, www.therobotreport.com/.